

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

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CSE 3100 : Software Development Lab

Project Proposal Report

Topic: Hotel Booking and Management System

Submitted by

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Roomify Hotel Management System

Project Report

1. Introduction

Roomify Hotel Management System is a modern, feature-rich web application developed to streamline hotel operations and enhance customer experience. The platform serves both hotel administrators and customers by offering efficient tools for room management, bookings, payments, staff handling, and content management through a centralized system.

The project focuses on usability, security, and scalability, ensuring that hotels can manage daily operations seamlessly while customers enjoy a smooth and intuitive booking journey.

2. Project Overview

Roomify is designed as an all-in-one hotel management solution with the following goals:

- Simplify hotel administrative operations
- Provide customers with an easy and secure booking experience
- Centralize room, staff, and customer management
- Offer a responsive and visually appealing interface

The system supports secure online payments, dynamic room availability checks, and role-based access for administrators.

3. Objectives

The primary objectives of the Roomify Hotel Management System are:

- To automate and simplify hotel management tasks
- To reduce manual booking errors and administrative overhead
- To enable customers to browse, filter, and book rooms effortlessly
- To empower administrators with full control over hotel resources
- To deliver a scalable and maintainable web-based solution

4. Key Highlights

- **Streamlined Booking System:** Easy room reservations with real-time availability and secure Stripe payments
 - **Powerful Admin Panel:** Full control over rooms, bookings, staff, testimonials, and website content
 - **User-Friendly Interface:** Responsive design compatible with desktops, tablets, and mobile devices
 - **Secure Transactions:** Integrated Stripe Payment Gateway for safe and reliable payments
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5. Technology Stack

Component	Technology
Backend	Laravel 8
Frontend	Bootstrap 5, jQuery 3
Database	MySQL
Payment Gateway	Stripe
Rendering	Server-Side Rendering
UI Design	Figma

6. System Features

6.1 Website Features

- **Home Page:** Highlights hotel services, rooms, and amenities
- **Gallery:** Visual display of rooms and hotel facilities
- **Services Page:** Detailed overview of available hotel services
- **About Us:** Information about the hotel's mission and values
- **Testimonials:**

- Customers can submit reviews after login
 - Admin approval required before public display
 - **Booking Page:**
 - Filter rooms by availability, capacity, and type
 - Select check-in and check-out dates
 - **Contact Page:** Allows customers to send direct inquiries
 - **FAQ Page:** Provides answers to common customer questions
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6.2 Admin Panel Features

- **Authentication:** Secure admin login and password recovery
 - **Content Management:**
 - Manage Home, Gallery, About Us, and FAQ sections
 - View and respond to customer inquiries
 - **Room Management:**
 - Add, edit, and delete rooms and room categories
 - Advanced room type and room number setup
 - **Booking Management:**
 - View, organize, and manage customer bookings
 - **Testimonial Management:**
 - Review and approve customer testimonials
 - **Staff Management:**
 - Maintain staff profiles and salary information
 - **Customer Management:**
 - Manage customer profiles
 - Track booking history and payments
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7. API Endpoints

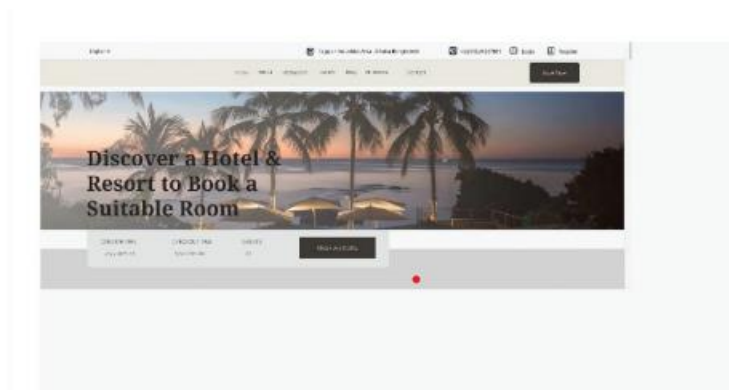
Method	Endpoint	Description
GET	/rooms	Fetch available rooms with filters
POST	/booking	Create a booking
POST	/testimonial	Submit a testimonial
GET	/staff	Retrieve staff details (Admin only)
POST	/payment	Process payments using Stripe

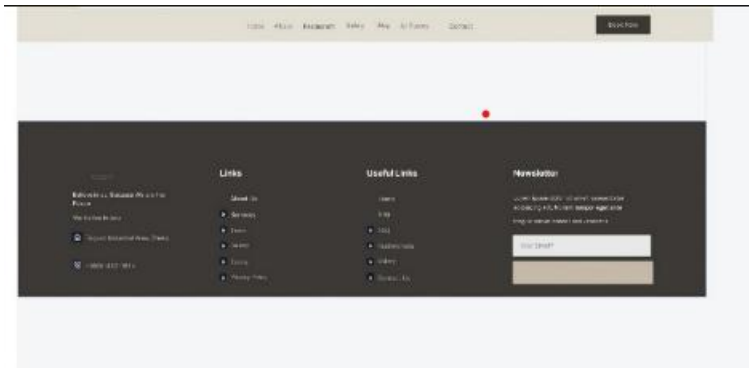
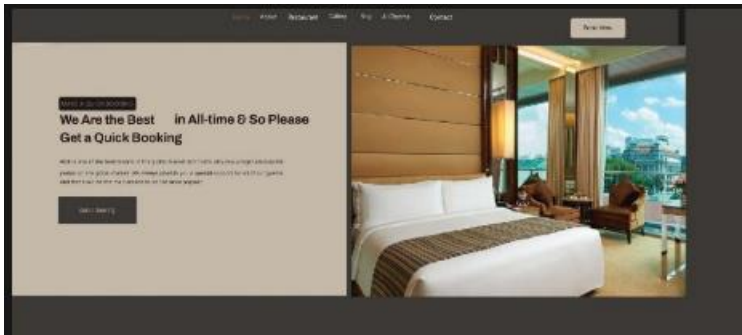
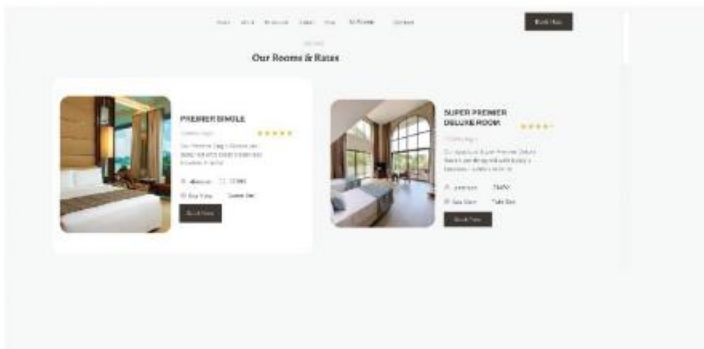
8. UI/UX Design

The user interface of Roomify was designed using **Figma and Canva**, focusing on:

- Clean and modern layout
- Consistent color schemes and typography
- Easy navigation for both customers and administrators
- Full responsiveness using Bootstrap 5

The design ensures a smooth user experience across all devices.





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Remember Me

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Sign In

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username

Link Facebook

Link Twitter

Change Password

Resetting Profile

Logout

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Accounting

0 Total

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Accounting

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Check In

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Check Out

00/00/0000

Number of Rooms

00

Number of Adults

00

Book Now



SUPER PREMIER DELUXE ROOM

Booking Details

Room No

Room No

Room No

Room No

Room No

Room No

Room No

Room No

Booking Summary



Room No

Room No

Room No

Room No

Room No

Room No

Room No

9. Project Milestones

Milestone 1: Foundation

- Laravel project setup and architecture
- User authentication (registration and login)
- Admin panel dashboard development

Milestone 2: Core Features

- Room type and advanced room setup
- Room number management
- Deleting rooms across related tables
- Display team members on frontend home page
- Booking area and room list implementation on frontend

Milestone 3: Final Touches

- Stripe payment gateway integration
- Booking filters and availability checks
- CRUD operations for rooms and staff
- Finalization of website content (Gallery, Services, FAQ)
- Testimonial submission and approval system
- Testing and deployment

10. Deployment & Environments

- **Preview Environment:** Used for testing and validation
 - **Production Environment:** Live deployment for end users
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11. Prerequisites & Setup

To run the project locally, the following are required:

- PHP >= 8.2
 - Composer
 - MySQL Database
 - Node.js and npm
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12. Contributors

- **Sazia Farah Deba (Frontend)**
 - **Tasnim Jarin (Backend + lead)**
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13. Conclusion

Roomify Hotel Management System delivers a comprehensive solution for modern hotel operations. By combining a powerful admin panel, intuitive customer interface, and secure payment processing, the platform significantly improves efficiency, customer satisfaction, and overall hotel management. The system is scalable, user-friendly, and well-suited for real-world hotel environments.

"An intuitive hotel management portal for seamless booking, room management, and customer experience."